From Quantum Shots to Training Data: Reorganizing Measurement Records in Quantum Machine Learning

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***Abstract*—Between keeping every shot as a noisy training example and averaging all shots into one clean feature lies an entire spectrum of data organizations that quantum machine learning usually leaves implicit. We make that choice explicit. Shot grouping partitions a fixed measurement record into disjoint groups and averages within each group, tuning smoothly between the two conventions with a single validated parameter and no additional quantum executions. We evaluate the method on chaotic synthetic benchmarks and on real-world data under strict execution budgets. At the balanced allocation, intermediate grouping lowers mean error across the evaluated tasks, with supported improvements on Santa Fe laser, Rössler, and sunspots. Matched controls show that the gain acts as correctly scaled regularization of the classical readout. Unmitigated runs on two superconducting processors show positive mean differences on every task. Shot grouping therefore offers a practical way to improve finite-shot quantum learning at no added quantum cost.**

***Index Terms*—Quantum computing, Machine learning, Reservoir computing, Time series analysis**

I. Introduction

Every quantum learning experiment ends the same way: a circuit runs many times, and each run returns one measured bitstring. Before anything can be learned, this pile of shots must become training data. The step looks like bookkeeping, and it is usually treated that way. Almost every pipeline averages all shots of a time step into a single expectation-value estimate and moves on. This paper asks what that habit costs.

The habit has a respectable origin. Expectation values are the native currency of quantum mechanics: they are what theory predicts and what experiments are built to estimate. A learning pipeline, however, does not consume observables. It consumes training examples, and it cares about two things only: how many there are and how noisy each one is. The moment a measurement record becomes training data, a physics convention quietly turns into a statistical decision: spend all measurements on making one example clean, or make several examples that are merely clean enough. No single choice can suit every task, budget, and noise level.

Averaging everything is one end of a dial. It buys the cleanest possible feature vector but pays with scarcity. Thousands of circuit executions collapse into a handful of training examples, and too few examples relative to the number of features destabilizes regression [1]. The opposite end, keeping every shot as its own example, buys abundance but pays with noise, and noisy features systematically shrink fitted coefficients [2]. Between these extremes lies an entire range of aggregations. Nothing in the record forces either convention, and the record does not care which one tradition prefers.

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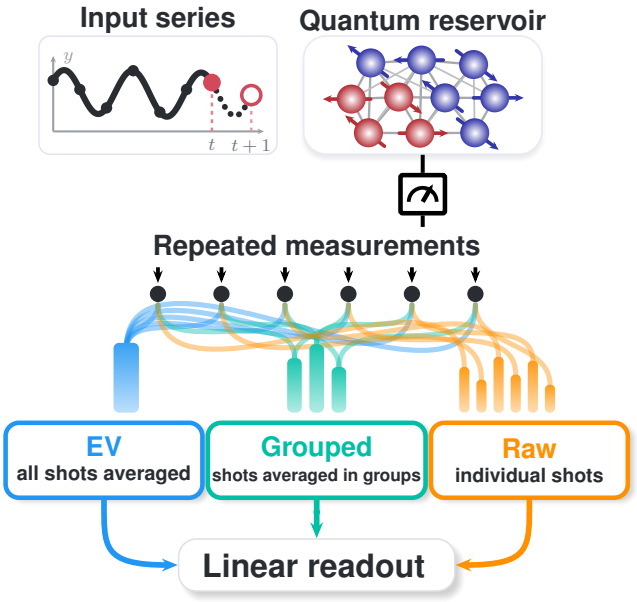


Fig. 1. Three ways to turn one shot record into training data. A fixed quantum reservoir is measured many times per step, and only the classical step changes: EV averages all shots, Raw keeps each shot, and Grouped averages disjoint groups. All feed the same linear readout.

Quantum reservoir computing (QRC) is a clean place to study this dial [3]–[5]. A fixed quantum circuit transforms an input sequence into measured features, and only a classical readout is trained. The circuit, the targets, and the measurement record can therefore be held completely fixed while the aggregation alone changes. Whatever happens to the forecast is attributable to the aggregation because nothing else moved.

We compare three representations of one record (Fig. 1). The expectation-value representation (EV) averages all measurements at each time step. Raw uses each measurement separately. Grouped averages disjoint subsets, and its aggregation scale k, the number of shots per averaged training view, slides smoothly from $k = 1$ (Raw) to $k = N$ (EV).

The dial turns out to matter. Under one matched protocol and fixed execution budgets, an intermediate aggregation lowers mean test error relative to full averaging on all four tasks at the balanced allocation. The improvement is supported on Santa Fe laser, Rössler, and sunspots, while Mackey–Glass is inconclusive. The best aggregation scale depends on the record, so no single convention is appropriate. The gain requires nothing quantum. Matched controls support a covariance-regularization explanation for the classical readout, supplied automatically by the record's own fluctuations. Because aggregation is classical post-processing, it costs no additional quantum executions and applies retroactively to any archived record.

Our contribution is *adaptive shot aggregation*. The number of shots per training view becomes a setting of the classical readout, selected by validation like any other hyperparameter, at no additional quantum cost. We map when the choice matters through three budget allocations of time steps and shots. We explain why it works through matched controls that isolate the regularization mechanism. We test where it transfers through unmitigated experiments with a separate hardware reservoir on two superconducting processors.

II. Related Work

Finite sampling can be addressed at three stages of a quantum learning pipeline: before execution, by allocating the measurement budget; after averaging, by correcting expectation-value estimates; or at the measurement record itself, by deciding how repeated outcomes become training examples. Our method focuses on this third stage.

A brief instance of subset averaging appears in Cemin *et al.* [6], where it is used as an implementation heuristic to enlarge the training set for reduced quantum channel learning. We develop this idea into a tunable procedure: the group size spans the full range from individual shots to complete expectation-value averaging, is selected jointly with readout regularization, and is evaluated under fixed execution budgets. For linear readouts, grouping also has an exact regularization interpretation, connecting it to classical work on training with noisy inputs [7].

Complementary work addresses the other stages of finite-shot learning. Resource-aware optimization allocates measurements before execution [8]. After averaging, noise-aware training rules [9] and variance regularization [10] reduce the effect of sampling noise, while fundamental limits for learning from finitely sampled physical systems have also been studied [11]. Shot grouping instead reorganizes an existing measurement record. It requires no additional quantum executions and can be applied retrospectively whenever individual outcomes have been retained.

Within QRC, prior work has examined how noise, dissipation, measurement, finite Hilbert-space dimension, and input transformations shape temporal features [12]–[23]. We hold the reservoir and measurement record fixed and change only how repeated outcomes are presented to the classical readout. Shot grouping is therefore complementary to these approaches and is not tied to a particular reservoir design.

III. Method

A. Adaptive shot aggregation

Write $x_t^{(n)}$ for the features produced by shot n at time t. We split the N shots of each step into $G = N/k$ groups of equal size k and average within each group,

$$x_t^{(g)} = \frac{1}{k} \sum_{n \in g} x_t^{(n)}, \qquad g = 1, \ldots, G. \tag{1}$$

Every group is a partly averaged view of the same time step, and all views share the same target y_t. At test time we average the G predictions of a step, which is identical to predicting once from the averaged features. The two methods are therefore scored in exactly the same way, so any difference between EV and Grouped is created during training, not at test time.

The shot index carries no physical meaning, because shot n at one step and shot n at the next come from unrelated circuit runs. We therefore shuffle the shots freshly at every time step before forming groups, and we repeat the whole analysis over several random groupings.

B. Fixed execution budget and training

Evaluating T time steps with N shots per step costs TN circuit executions. We hold this budget fixed at ten thousand executions and study three ways of spending it, $(T, N) = (250, 40)$, $(500, 20)$, and $(1000, 10)$, ranging from fewer time steps with more shots to more time steps with fewer shots. For each setting we try every group size k that divides N, which includes Raw at $k = 1$ and EV at $k = N$.

Selection and testing respect time order. After a short washout, the earlier part of each series is the selection data used to choose settings, and the final fifth is held out as the test block and left untouched until the end. On the selection data we pick the group size and ridge strength (k, λ) by forward-chaining validation. In three successive rounds we fit on an earlier stretch and score on the stretch that immediately follows, so training always precedes validation in time, and each candidate pair is ranked by its validation error averaged over the three rounds. The best pair is then frozen, the readout is refitted on the whole selection data, and the test block is scored once. Normalization and feature scaling are computed from training data only, so the test block never influences any choice.

C. Why grouping regularizes the readout

Full averaging gives the readout one relatively stable feature vector per time step. Raw gives it many noisier vectors. Grouping lies between them, providing several partly averaged views of the same time step with the same target.

Stable feature combinations should predict similarly across these views, whereas shot-sensitive combinations do not. Fitting every view to the same target therefore discourages reliance on finite-shot fluctuations.

For a linear readout with squared loss, this effect is exact. Let $\bar{x}_t$ be the average of the grouped views, and let $C_{k,t} =$

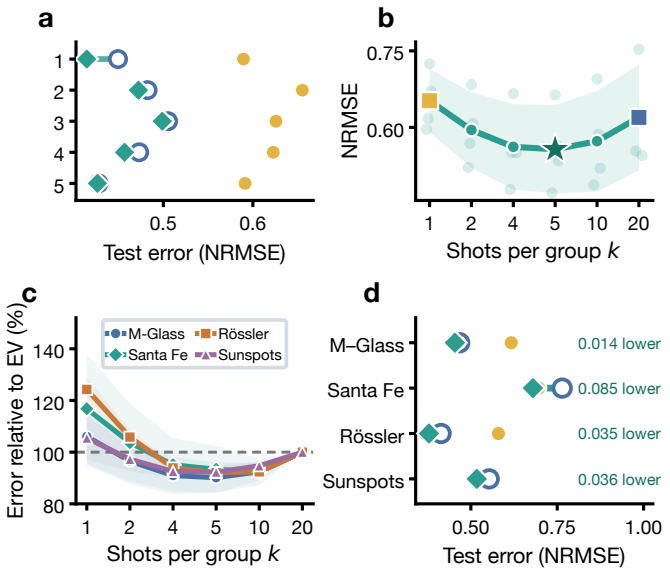


Fig. 2. Adaptive aggregation under the shared protocol. (a) Grouped (teal), EV (open circle), and Raw (yellow) on five Mackey–Glass runs. (b) The error is U-shaped in the aggregation scale k, lowest at an intermediate value (star is the validated choice). (c) The same dip below EV appears in the task means. (d) After tuning, Δ shows the paired mean difference between EV and Grouped for each task. Panels (b,c) use development data, (a,d) test blocks.

$\frac{1}{G}\sum_g (x_t^{(g)} - \bar{x}_t)(x_t^{(g)} - \bar{x}_t)^\top$ be their covariance around that average. For any intercept b and weights w,

$$\frac{1}{G}\sum_{g=1}^{G}\left(y_t - b - w^\top x_t^{(g)}\right)^2 = \left(y_t - b - w^\top \bar{x}_t\right)^2 + w^\top C_{k,t} w. \quad (2)$$

The first term is the EV loss. The second penalizes feature combinations that fluctuate across views. Grouped training is therefore EV training with a regularizer obtained from observed shot fluctuations. The identity requires neither Gaussian noise nor independent time steps.

On a fixed common feature scale, let $S_{\mathrm{shot},t}$ be the $1/N$ covariance of the N shot-level features. Before representation-specific standardization, averaging over random partitions $\mathcal{P}$ gives

$$\mathbb{E}_{\mathcal{P}}[C_{k,t}] = \frac{N-k}{k(N-1)} S_{\mathrm{shot},t}. \quad (3)$$

The multiplier is one at $k = 1$ and zero at $k = N$. Thus, k mainly controls regularization strength: smaller groups retain more shot variation, while larger groups average it away. A finite partition can also change the realized penalty, especially when only a few groups are available.

This motivates covariance-regularized expectation-value training (Cov-EV), which retains EV features and adds a penalty estimated from shot fluctuations. Its covariance and ridge strengths are selected on the same validation data, independently of Grouped's k. Similar performance from Grouped and Cov-EV would support covariance regularization as the mechanism behind the gain.

As a check on our implementation, at a fixed group size and ridge strength we compute the Grouped fit in two mathematically equal ways. The first fits ridge directly on all grouped training views. The second uses the two-term form of Eq. (2), fitting on the mean view with the added covariance penalty. With the same feature standardization, the two agree to numerical precision, which confirms that the decomposition is implemented correctly.

IV. Experimental Setup

We run two studies. The first is a controlled simulation on four forecasting tasks. The second repeats the core comparison on real quantum hardware. In both, every aggregation choice reads the very same measurement record, so any difference in accuracy comes only from how the shots are organized and not from running the circuit differently.

A. Simulation reservoir and evaluation protocol

The simulated reservoir is a fixed six-qubit circuit with two shallow data reuploading layers [24]. Each layer applies input-dependent R_y and R_z rotations with fixed per-qubit offsets, nearest-neighbor R_{zz} couplings, and fixed non-Clifford mixing rotations. For each circuit setting, all angles are sampled once and then held fixed; three frozen settings are used to assess sensitivity to this draw. Each computational-basis shot yields 11 features: six local observables Z_i and five nearest-neighbor products Z_iZ_{i+1}. A leaky integrator with leak rate 0.2 and a ten-step lag window provide temporal memory, and a ridge-regression readout predicts the next value.

We evaluate one-step-ahead forecasting on four time series: the synthetic Mackey–Glass [25] and Rössler [26] systems, and the measured Santa Fe laser intensity [27] and monthly sunspot number [28]. All tasks use the same reservoir architecture, feature construction, temporal processing, scaling, validation procedure, and execution budgets. The same selection rule is applied separately to each task; no task-specific changes are made to the reservoir or preprocessing pipeline. Santa Fe is a later exploratory extension under the same protocol and is not a preregistered confirmation.

EV, Grouped, Raw, and Cov-EV are derived from the same finite-shot measurement record. EV averages all shots at each time step, Raw retains individual shots, Grouped averages shots within disjoint groups, and Cov-EV uses the fully averaged EV features with a penalty estimated from shot-level covariance. Two additional controls use different feature sources: an ideal reservoir without finite-shot sampling noise and a classical baseline that applies the same temporal processing and readout directly to the input series. These controls distinguish the effect of aggregation from the effects of finite sampling and of the reservoir transformation itself.

The trajectory or time window is the unit of uncertainty estimation. Each synthetic task uses five independently generated trajectories, and each real-world task uses five windows with disjoint test blocks. Within each unit, we repeat the analysis over three circuit settings, three measurement draws, and three random groupings where applicable. These repeats are averaged within the unit before paired confidence intervals are formed across the five units. Grouped views, neighboring time steps, and repeated evaluations of the same sequence are not treated

as independent observations, because they share measurements or are coupled by the leaky integrator and overlapping lags.

Model selection uses no test data. After a short washout, the final 20% of each trajectory or window is reserved as an untouched test block. The preceding data are used in three expanding-window validation splits, each trained on the available past and evaluated on the immediately following block. The group size k and ridge strength λ are selected from these validation results. The readout is then refitted on all data preceding the test block and evaluated once on the reserved block. For representations with multiple views at one time step, the corresponding predictions are averaged to produce one forecast, so all methods are scored on the same targets.

The primary budget uses $T = 500$ time steps and $N = 20$ shots per step, giving 10,000 circuit executions. We also test $(T, N) = (250, 40)$ and $(1000, 10)$, which spend the same budget with different balances between observed time steps and shots per step. All reservoir parameters, data-generation settings, window origins, and selection rules are fixed and stored before any test block is evaluated.

B. Hardware setup

The hardware study deliberately uses a different reservoir. It is a fixed ten-qubit circuit with one input qubit, nine memory qubits, a single ring of entangling gates, and ten local outputs, mapped to selected qubits on the *ibm_fez* and *ibm_marrakesh* processors and held fixed across all four tasks. Its purpose is to check whether the same aggregation procedure still helps on a second circuit and a different set of measured features. It is not a hardware implementation of the six-qubit simulator and not a comparison of qubit counts. As in simulation, every method receives the same circuit outputs and only the classical grouping of shots changes. For each task and processor we take one acquisition of 400 shots at each of 500 time steps and split it into twenty ordered, disjoint records of 20 shots. The resulting intervals describe how much the result varies among the measurement blocks of that single acquisition. They do not stand in for independent devices or separate calibration periods.

V. Evaluation

We report accuracy as test NRMSE, a normalized error for which lower is better and where one means predicting the mean. Three ways of turning the shot record into features are compared throughout. EV averages all shots of a step into one clean feature vector. Raw keeps every shot as its own noisy example. Grouped is our method and sits between them. A fourth entry, Cov-EV, is a control that adds an independently tuned covariance penalty to plain averaging. We summarize the benefit of Grouped as the paired difference EV minus Grouped, so a positive value means Grouped is more accurate. Each difference comes with a 95% confidence interval. An interval above zero provides evidence of a positive mean improvement; an interval containing zero is inconclusive. The balanced setting spends the fixed budget on 500 time steps with 20 shots each.

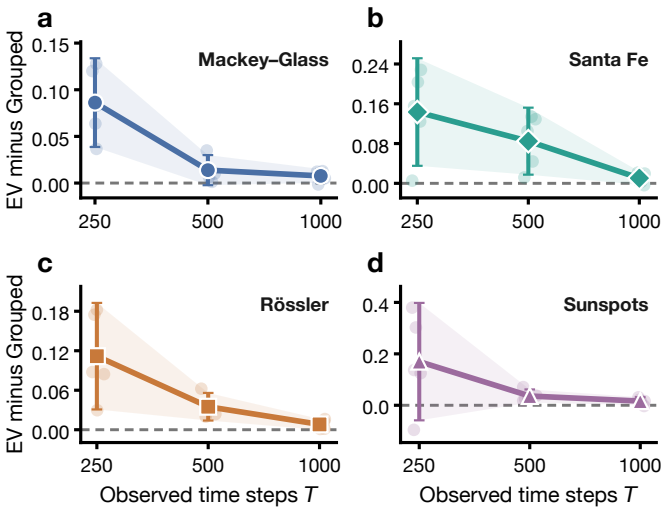


Fig. 3. When aggregation helps most, at a fixed budget. Each panel shows the improvement of Grouped over EV, measured as $\mathrm{NRMSE}_{\mathrm{EV}} - \mathrm{NRMSE}_{\mathrm{Grouped}}$, so values above zero favor Grouped. Moving right spends the budget on more steps and fewer shots. The gain is largest with many shots and few steps. Markers and bars are means with 95% intervals.

TABLE I
Mean test NRMSE at the balanced setting. Repeated measurement trials are averaged within each of five separate trajectories or windows. Lower values are better.

Task	Class.	Ideal	EV	Grouped	Raw	Cov-EV
Mackey–Glass	0.100	0.178	0.468	0.454	0.617	**0.447**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

A. Balanced setting

At the balanced budget, Grouped lowers mean NRMSE relative to EV on all four tasks (Table I, Fig. 2). The improvement is 0.0138 on Mackey–Glass (95% CI $[-0.0023, 0.0300]$), 0.0848 on Santa Fe laser ($[0.0174, 0.1522]$), 0.0349 on Rössler ($[0.0139, 0.0559]$), and 0.0360 on sunspots ($[0.0103, 0.0616]$). The interval stays above zero on the two real-world datasets, Santa Fe laser and sunspots, and on Rössler. Only Mackey–Glass is inconclusive.

Santa Fe laser shows the largest effect. Grouped reduces the error from 0.764 to 0.680, and all five disjoint test windows favor Grouped. When k and the ridge strength are tuned together, the search lands on $k = 10$ most often and on an intermediate scale, neither Raw nor EV, in 94% of repeated trials. Cov-EV reaches 0.677, almost the same as Grouped, which already suggests that the gain is a regularization effect rather than new physical information. The gain also depends on how the budget is spent. On Santa Fe it shrinks from 0.143 with 40 shots per step to 0.010 with 10 shots per step because grouping has more shots to reorganize when shots are plentiful.

We also compare an ideal noise-free reservoir with a classical baseline that uses identical preprocessing. The quantum reservoir is worse on Mackey–Glass and sunspots, better on Santa Fe, and inconclusive on Rössler. Our claims are therefore about organizing a chosen reservoir record, not quantum advantage.

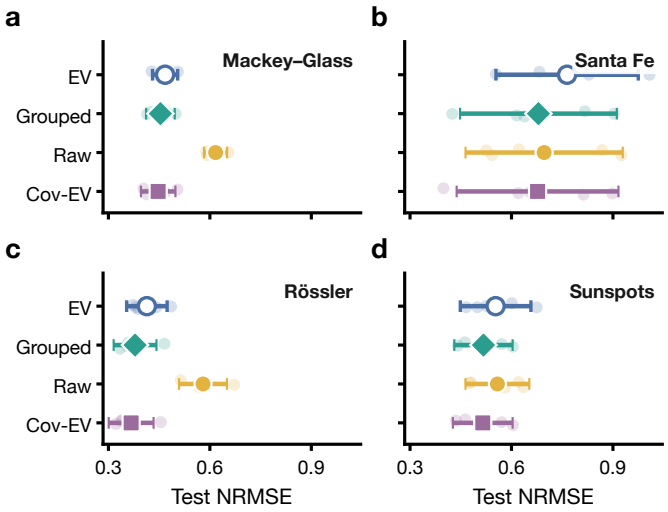


Fig. 4. Controls at the balanced setting, lower is better. On every task Grouped (teal diamond) and Cov-EV (purple square) are lowest and nearly equal, while EV (open circle) and Raw (yellow) trail. Their near-match supports a readout-regularization explanation. Faint points are the five unit means, markers and bars are means with 95% intervals.

The preferred aggregation scale varies across tasks, trajectories, circuit settings, measurement draws, and grouping assignments, and EV is occasionally best. On a fixed feature scale, Eq. (3) clarifies the role of k: it mainly sets the strength of the covariance penalty, while a finite random grouping can also change the realized penalty. Raw is less accurate than Grouped on all four task means.

B. Trading time steps against shots

Figure 3 compares three ways of spending the fixed budget. With 40 shots at each of 250 time steps, Grouped improves every task by 0.086 to 0.170. The interval stays above zero for Mackey–Glass, Santa Fe laser, and Rössler, while the sunspot result varies across windows. When the same budget is spread over 1000 time steps with only 10 shots each, the gains shrink to between 0.008 and 0.016, and the interval stays above zero only for Mackey–Glass and sunspots.

C. Why the gain is regularization

Cov-EV keeps EV features and adds a penalty based on how the shots fluctuate together. At the balanced setting its mean error is 0.0021 to 0.0116 NRMSE below Grouped, and only the Rössler interval excludes zero. This near-match supports the regularization mechanism in Eq. (2) rather than extra device information (Fig. 4). Raw is less accurate than EV on Mackey–Glass and Rössler and inconclusive on the two real-world datasets. At fixed k, the exact two-term loss reproduces the Grouped ridge fit to numerical precision. On a fixed feature scale, Eq. (3) shows how the induced penalty changes with k, whereas Cov-EV tunes the full shot-covariance penalty directly.

D. Transfer to hardware

Figure 5 reports the unmitigated hardware results. Grouped improves every task, with mean gains from 0.038 to 0.153 NRMSE on *ibm_fez* and from 0.031 to 0.103 on *ibm_marrakesh*. On Santa Fe laser, the error falls from 0.750 to 0.711 on *ibm_fez* and from 0.744 to 0.713 on *ibm_marrakesh*. Grouped is more accurate in 19 of 20 measurement blocks on each processor.

VI. Discussion

A quantum experiment produces a measurement record, not a ready made training set. Combining repeated shots into feature vectors is therefore part of the learning algorithm. EV averages all N shots at each step. Adaptive shot aggregation instead averages groups of size k, from Raw at $k = 1$ to EV at $k = N$, without changing the circuit or record. The settings are fixed before testing, and gains arise during training at no additional quantum cost.

The choice matters most when many shots are collected at relatively few time steps. Full averaging then spends a large measurement budget on a small number of highly precise examples. Grouping creates moderately denoised views from the same shots. Their differences reveal which feature directions are stable under finite shot fluctuations. With fewer shots per step, the freedom to reorganize the record and the benefit both shrink. The aggregation scale should therefore be validated rather than fixed by convention.

For a linear readout, this effect has an exact interpretation. Equation (2) separates the grouped loss into the EV loss and a quadratic covariance penalty. Since every view shares the same target, the penalty discourages large weights on feature combinations that vary between groups. On a common feature scale, Eq. (3) shows that smaller groups produce stronger regularization and $k = N$ recovers EV.

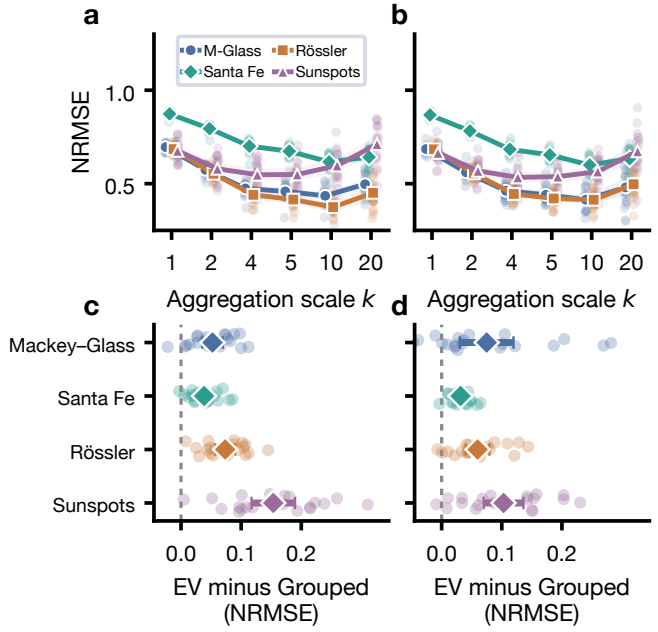


Fig. 5. Transfer to two unmitigated processors. (a,b) On *ibm_fez* and *ibm_marrakesh* the error is again U-shaped in the aggregation scale k. (c,d) The improvement of Grouped over EV is positive on all four tasks and both devices. One $T = 500$ acquisition of 400 shots per step is split into twenty disjoint $N = 20$ records, and intervals describe variation within that single acquisition.

The Cov EV control supports this explanation. It keeps the fully averaged features but adds a tuned covariance penalty estimated from shot fluctuations. It stays close to Grouped on every task and reaches slightly lower mean error overall, pointing to a shared mechanism. Cov EV tunes the penalty directly, while Grouped uses strengths produced by the available group sizes. Grouping is therefore a simple route to this regularization, but not the only one.

Individual shot outcomes should be retained. Once a record has been reduced to expectation values, alternative aggregation scales cannot be reconstructed. Keeping the full record preserves this option without repeating the quantum experiment.

The effect transfers to a separate ten qubit circuit run without mitigation on two superconducting processors, where Grouped improves on EV across all tasks. The hardware intervals measure blocks within one acquisition per task and processor, not reproducibility across runs, calibration states, or days. Independent hardware campaigns, other reservoirs, and richer readouts are the next tests. More broadly, finite shot processing is part of the learning algorithm.

VII. Conclusion

Turning a shot record into training data has always involved a choice, and the field has usually answered it by averaging everything. Choosing the aggregation scale by validation lowers mean test error on all four forecasting tasks at the balanced budget. The improvement is supported on Santa Fe laser, Rössler, and sunspots, while Mackey–Glass is inconclusive. The hardware study shows positive mean differences on both superconducting processors, and the method requires no additional quantum cost. Because every measurement-based reservoir produces such a record, the choice is available to all of them. The underlying question, how to turn finitely sampled noisy measurements into training data, reaches beyond reservoirs to finite-shot learning as a whole.

References

[1] D. Hsu, S. M. Kakade, and T. Zhang, "Random Design Analysis of Ridge Regression," in *Proceedings of the 25th Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 23, 2012, pp. 9.1–9.24.

[2] W. A. Fuller, *Measurement Error Models*. New York: Wiley, 1987.

[3] K. Fujii and K. Nakajima, "Harnessing Disordered-Ensemble Quantum Dynamics for Machine Learning," *Physical Review Applied*, vol. 8, no. 2, p. 024030, 2017.

[4] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, "Boosting Computational Power through Spatial Multiplexing in Quantum Reservoir Computing," *Physical Review Applied*, vol. 11, no. 3, p. 034021, 2019.

[5] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi *et al.*, "Opportunities in Quantum Reservoir Computing and Extreme Learning Machines," *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.

[6] G. Cemin, M. Cech, E. Weiss, S. Soltan, D. Braun *et al.*, "Machine Learning of Reduced Quantum Channels on Noisy Intermediate-Scale Quantum Devices," *Physical Review A*, vol. 110, no. 5, p. 052418, 2024.

[7] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[8] C. Moussa, M. H. Gordon, M. Baczyk, M. Cerezo, L. Cincio, and P. J. Coles, "Resource Frugal Optimizer for Quantum Machine Learning," *Quantum Science and Technology*, vol. 8, no. 4, p. 045019, 2023.

[9] O. Ahmed, F. Tennie, and L. Magri, "Optimal Training of Finitely Sampled Quantum Reservoir Computers for Forecasting of Chaotic Dynamics," *Quantum Machine Intelligence*, vol. 7, p. 31, 2025.

[10] D. A. Kreplin and M. Roth, "Reduction of Finite Sampling Noise in Quantum Neural Networks," *Quantum*, vol. 8, p. 1385, 2024.

[11] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci *et al.*, "Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks," *Physical Review X*, vol. 13, no. 4, p. 041020, 2023.

[12] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Quantum Noise-Induced Reservoir Computing," 2022.

[13] A. Palacios de Luis, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Role of Coherence in Many-Body Quantum Reservoir Computing," *Communications Physics*, vol. 7, no. 1, p. 369, 2024.

[14] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Temporal Information Processing Induced by Quantum Noise," *Physical Review Research*, vol. 5, no. 2, p. 023057, 2023.

[15] R. Martínez-Peña and J.-P. Ortega, "Quantum Reservoir Computing in Finite Dimensions," *Physical Review E*, vol. 107, no. 3, p. 035306, 2023.

[16] P. Mujal, R. Martínez-Peña, G. L. Giorgi, M. C. Soriano, and R. Zambrini, "Time-Series Quantum Reservoir Computing with Weak and Projective Measurements," *npj Quantum Information*, vol. 9, p. 16, 2023.

[17] Y. Suzuki, Q. Gao, K. C. Pradel, K. Yasuoka, and N. Yamamoto, "Natural Quantum Reservoir Computing for Temporal Information Processing," *Scientific Reports*, vol. 12, p. 1353, 2022.

[18] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Dissipation as a Resource for Quantum Reservoir Computing," *Quantum*, vol. 8, p. 1291, 2024.

[19] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, and T. A. Ohki, "Nonlinear Input Transformations Are Ubiquitous in Quantum Reservoir Computing," *Neuromorphic Computing and Engineering*, vol. 2, no. 1, p. 014008, 2022.

[20] R. Martínez-Peña, J. Nokkala, G. L. Giorgi, R. Zambrini, and M. C. Soriano, "Information Processing Capacity of Spin-Based Quantum Reservoir Computing Systems," *Cognitive Computation*, vol. 15, no. 5, pp. 1440–1451, 2023.

[21] W. D. Kalfus, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, T. A. Ohki, and L. C. G. Govia, "Hilbert Space as a Computational Resource in Reservoir Computing," *Physical Review Research*, vol. 4, no. 3, p. 033007, 2022.

[22] D. Fry, A. Deshmukh, S. Y.-C. Chen, V. Rastunkov, and V. Markov, "Optimizing Quantum Noise-Induced Reservoir Computing for Nonlinear and Chaotic Time Series Prediction," *Scientific Reports*, vol. 13, p. 19326, 2023.

[23] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, and T. A. Ohki, "Quantum Reservoir Computing with a Single Nonlinear Oscillator," *Physical Review Research*, vol. 3, no. 1, p. 013077, 2021.

[24] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.

[25] M. C. Mackey and L. Glass, "Oscillation and Chaos in Physiological Control Systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977.

[26] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, 1976.

[27] N. A. Gershenfeld and A. S. Weigend, "The Future of Time Series," Santa Fe Institute, Tech. Rep. 93-08-053, 1993.

[28] SILSO World Data Center, "Monthly mean total sunspot number V2.0," 2026.